



Improving the accuracy of teachers' judgments of student learning

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HIGHLIGHTS

- Professional development in student-centered instruction improved the accuracy of teachers' judgments of student learning.
- Professional development on formative assessment did not affect the accuracy of teachers' judgments.
- Accuracy of teacher judgments were associated with student achievement.
- Professional development in student-centered instruction of mathematics also improved student achievement.
- Relative accuracy of teacher judgments mediated the effect of professional development on student achievement.

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ABSTRACT

We examined the effect of different professional development programs on the accuracy of teachers' judgments of their students' learning during three academic years. Teachers participated in a program focused on improving (a) use of formative assessment, (b) student-centered mathematics instruction, (c) use of both formative assessment and student-centered mathematics instruction; or (d) neither—a control group. Teachers' judgment accuracy was greater for teachers who participated in professional development around improving student-centered mathematics instruction than for other groups. A multilevel analysis showed a significant positive relation between the accuracy of teachers' judgments and student achievement.

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1. Introduction

Models of instruction describe teaching as the interplay between teachers' judgments of student learning and subsequent instruction (e.g., Box, Skoog, & Dabbs, 2015; Donovan, Bransford, & Pellegrino, 2000; Ready & Wright, 2011; Ruiz-Primo & Furtak, 2007). That is, teachers' judgments of student learning guide instruction (e.g., Alvidrez & Weinstein, 1999). For instance, judgments of student learning help teachers identify struggling students for additional instruction (e.g., Bailey & Drummond, 2006), and shaping feedback to students (e.g., Hoge & Coladarsi, 1989). Given the role of teacher judgments in instruction, it is important to find ways to improve the accuracy of teachers' judgments. The present investigation evaluates the efficacy of different professional development programs on teachers' judgment accuracy.

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Research on teacher judgments has been conducted in a number of countries. Südkamp, Kaiser, and Möller (2012) noted in their recent review of the teacher judgment literature that a substantial amount of research has focused on how the accuracy of teacher judgments is affected by the characteristics of the judgment, test, students, and teachers (see also Hoge & Coladarsi, 1989). This research, for instance, has shown that judgment accuracy is better when teachers are informed about the nature of the test prior to judging student performance than when they are uninformed (e.g., Feinberg & Shapiro, 2009) and when tests are more closely related to the classroom curriculum than when they are more general standardized tests (e.g., Jenkins & Demaray, 2016). It has also shown that judgment accuracy is better when teachers judge performance of higher achieving students than lower achieving students (e.g., Hurwitz, Elliot, & Braden, 2007), and judgment accuracy is not influenced by teaching experience (Zhu & Urhahne, 2015). However, relatively little research has examined ways to improve judgment accuracy—which is the focus of the present study. Before reviewing the literature focused on improving judgment accuracy, we first describe how judgment accuracy has been measured and

operationalized in the literature.

1.1. Measuring judgment accuracy

The standard procedure for gathering teacher judgments involves showing teachers a test and asking them to predict how their students will do on the test. Students then complete the test. Judgment accuracy is operationalized as the match between predicted and actual performance.

Judgment accuracy that describes the degree to which the magnitude of judgments is related to the actual magnitude of students' performance on a test is called *absolute accuracy* (Dunlosky & Rawson, 2012) and is often reported as confidence bias (i.e., average predicted performance minus average actual performance computed across students). Teachers typically over estimate students' performance, which is reflected in positive confidence bias (e.g., Rausch, Karing, Dörfler, & Artelt, 2016). For instance, over half the teachers in Bates and Nettelbeck (2001) were over-confident by 9–12 months on standardize reading assessment. *Relative accuracy* describes the degree to which judgments discriminate between different levels of performance across students. This is typically reported as the intra-teacher correlation between predicted and actual performance computed across students (e.g., Helmke & Schrader, 1987). Relative accuracy varies dramatically across studies. Accuracy reported in the meta-analysis conducted by Südkamp et al. (2012) varied from $r = 0.80$ (Methe, Hintze, & Floyd, 2008) to $r = -0.03$ (Graney, 2008).

It is important to note that absolute accuracy and relative accuracy are statistically independent of one another (Dunlosky & Thiede, 2013). To illustrate this point, we have created data for four teachers, each of whom is predicting the performance of five students on a test with 10 questions. As seen in Table 1, Teacher 1 is neither over-confident nor under-confident because the confidence bias is zero; this represents perfect absolute accuracy. Teacher 1 also has perfect relative accuracy—as the correlation between predicted and actual performance computed across the five students is $+1.0$. Teacher 2 is over-confident by an average of five points, but the correlation between predicted and actual performance is $+1.0$; thus, Teacher 2 has poor absolute accuracy, but perfect relative accuracy. Teacher 3 has perfect absolute accuracy (as the average predicted performance minus the average actual performance equals zero), but has relative accuracy of -1.0 (which is perfectly inaccurate). Teacher 4 has poor absolute accuracy and poor relative accuracy.

To illustrate the link between judgment accuracy and instruction, consider instructional decisions for these four teacher. Regarding absolute accuracy, Teachers 2 and 4 are quite over-confident. They think their students will get, on average, seven of ten points on the test, which may be well enough for the teacher to

proceed to the next topic. However, this overconfidence may lead the teachers to advance before their students have adequately learned the materials; therefore, foundational ideas may remain unlearned. By contrast, Teachers 1 and 3 may pace instruction more appropriately because their judgments accurately reflect their students' performance. Regarding relative accuracy, Teachers 3 and 4 did not accurately discriminate the students who have learned more from those who have learned less. These teachers might allocate more time to work individually with Students 4 and 5 (who actually did better on the test) rather than work with Students 1 and 2 (who actually did worse on the test). By contrast, Teachers 1 and 2 had perfect relative accuracy. The students they judged to have better learned the materials in fact did better on the test and the students they judged to have less mastery of the materials in fact did worse on the test; therefore, they are more likely to work with students who would benefit from additional instruction. Thus, judgment accuracy can influence decisions about pacing of instruction (absolute accuracy) and decisions about tailoring instruction to the needs of individual students (relative accuracy) (Thiede, Oswalt, Brendefur, Carney, & Osguthorpe, in press).

An alternative to asking teachers to predict performance on the test as a whole is to have teachers predict performance item by item for each student (for a discussion see Coladarsi, 1986). This approach makes it possible to compute accuracy as a hit rate across individual items. A teacher with a high hit rate can accurately judge how a student will do on each item. Knowing how each student would perform on each item would allow a teacher to tailor instruction to the item level.

1.2. Improving judgment accuracy

The cue-utilization framework (Koriat, 1997) provides a model for understanding how to improve monitoring accuracy. According to the framework, monitoring accuracy is driven by the cues people use to make judgments. Monitoring accuracy should improve when people have access to cues that are diagnostic of subsequent test performance (e.g., Thiede, Griffin, Wiley, & Anderson, 2010; Hertzog, Dunlosky, Robinson, & Kidder, 2003) and then use these cues to make judgments (Koriat & Bjork, 2005; Van Loon, de Bruin, van Gog, van Merriënboer, & Dunlosky, 2014). The present investigation examined how judgment accuracy was affected by two approaches used to increase access and use of diagnostic cues.

Use of formative assessment. (This section is now before the section on student-centered instruction.) Past performance is often the best statistical predictor of future performance. Teacher judgments are influenced by students' past performance (Hecht & Greenfield, 2002; Martínez, Stecher, & Borko, 2009). That is, teachers may use scholastic history in general (Dusek & Joseph, 1983) or performance on formative assessments conducted

Table 1
Illustrations of the relation between absolute and relative accuracy.

Performance	Teacher 1		Teacher 2		Teacher 3		Teacher 4	
	Predicted	Actual	Predicted	Actual	Predicted	Actual	Predicted	Actual
Student 1	9	9	9	4	9	5	9	0
Student 2	8	8	8	3	8	6	8	1
Student 3	7	7	7	2	7	7	7	2
Student 4	6	6	6	1	6	8	6	3
Student 5	5	5	5	0	5	9	5	4
Average	7	7	7	2	7	7	7	2
Accuracy								
Absolute	0		+5		0		+5	
Relative	+1.0		+1.0		-1.0		-1.0	

during instruction (Thiede et al., in press) to judge student learning. The diagnosticity of past performance will depend on how well past performance is aligned to subsequent tests. When learning objectives, instruction, and assessments are well aligned, formative assessments should provide highly diagnostic cues for predicting performance on summative assessments (Stiggins & Chappuis, 2006; Stiggins, 2008). There is evidence that use of formative assessment is positively related to the accuracy of teachers' judgments of student learning. For instance, Thiede et al. (in press) showed that the frequency with which teachers actively monitored of student learning and provided feedback to students during instruction were positively correlated with judgment accuracy. However, no published research has examined whether professional development to increase use of formative assessment during instruction improves judgment accuracy.

Student-centered instruction. One way to increase teachers' focus on student learning during instruction is to encourage them to use student-centered instructional techniques (Connor et al., 2014; Curby, Rimm-Kaufman, & Cameron Ponitz, 2009). Use of student-centered instructional techniques promote conversations about learning among students and between students and the teacher. For instance, at the core of *Cognitively Guided Instruction (CGI)* is the teacher practice of listening to students about their mathematical thinking and using this to guide instruction (Carpenter, Fennema, Peterson, Chiang, & Loef, 1989; Peterson, Carpenter, & Fennema, 1989). Evaluation of the efficacy of CGI professional development has included assessing the accuracy with which teachers judge students' thinking. In particular, teachers were asked to predict the strategies students would use to solve problems and whether students would correctly solve each problem. Although CGI did not lead to more accurate predictions regarding strategy use, it improved the accuracy of predictions regarding correct problem solving—absolute accuracy (Carpenter et al., 1989; Gabriele, Joram, & Park, 2016; Peterson et al., 1989).

Thiede et al. (2015) used the *Developing Mathematical Thinking (DMT)* professional development (Brendefur, Carney, Hughes & Strother, 2015; Brendefur, Thiede, Strother, Jesse, & Sutton, 2016) to improve student-centered instruction. At the core of DMT is the idea that teachers facilitate conversations about students' mathematical thinking—it has been shown to increase teachers' perceived level of student-centered teaching (Carney, Brendefur, Hughes, & Thiede, 2015; Carney, Brendefur, & Thiede, 2016). Thiede et al. (2015) showed DMT improved the relative accuracy of teachers' judgments of student learning. Although the DMT professional development focuses on promoting student-centered instruction, it does not include professional development specifically targeting use of formative assessment.

In the present study, we examined the effect of different professional development programs on the accuracy of teachers' judgments of student learning. Using a factorial design, teachers either received professional development to improve their use of student-centered instruction (DMT) or not; and teachers either received professional development to improve their use of formative assessment (FA) or not. Thus, we had a 2 (DMT: Yes versus No) $\times$ 2 (FA: Yes versus No) design.

1.3. Overview of study and professional development

As part of this grant-funded project, data collection for this study took place over three years. We initially recruited schools by meeting with school principals and describing the study. The principals of all the elementary schools in the district were interested in participating in the study. We worked with the mathematics coordinator of the district to create two sets of four schools, for each year, matched on (a) prior performance on the

standardized test of mathematics, (b) proportion of students eligible for free or reduced-price lunch, and (c) proportion of ELL students. Thus, for a given year, prior mathematics performance, proportion of students eligible for free or reduced-price lunch, and proportion of ELL students were equivalent within each set of schools, but differed across the two sets. Principals of these schools affirmed their interest in participating in the study and selected two to three teachers to participate per grade (grades K through 5). We did not dictate how principals selected teachers for participation; therefore, we cannot rule out that principals used different criteria for selecting teachers for participation—which is a limitation to the current study. However, it is important to note that at the time of selection, principals were blind to the group their teachers would be assigned, which means principals could not systematically bias the results of the study by selecting teachers particularly well suited for a specific professional development program or the control group. It is also important to note that teachers initially selected to participate completed the study; thus, there was no systematic bias created by attrition from the study. Randomly assigning schools to condition should have produced groups of teachers with equivalent judgment accuracy prior to professional development—we did not gather pretest data to verify equivalence because the professional development began in the summer. We used the same process for recruiting teachers each year of the three year project.

For teachers who received professional development, the structure of the professional development was held constant across the treatment groups. It included a three-day workshop during the summer and two 4-h after-school workshops, which were conducted early in the school year to provide more support early and then reduced scaffolding over the year. Professional development also included extensive embedded classroom support that included grade-level team unit planning sessions, co-teaching of lessons, and conversations around best practice. Across unit planning and individual sessions, we worked with teachers approximately 20 times during the year. The professional development involved working with teachers on three units. The first unit planning sessions occurred at the beginning of the school year in order to provide intensive assistance early, and to allow these early units to serve as models for units to be developed later in the year. Below are descriptions of the core features and structure of each professional development program.

1.3.1. Formative assessment (FA) professional development

FA Core Features. FA is an iterative process designed to obtain information about student understanding to inform subsequent instruction. The specific components of instruction that integrate high-quality FA are (1) identification of learning objectives, (2) monitoring student learning towards objectives, (3) providing feedback, and (4) adjusting instruction based on the teachers' judgments of student progress towards the learning objectives (e.g., Fulmer, 2011; Murphy & Holme, 2014). From the perspective of providing professional development targeted at improving teachers' integration of FA, it must focus on assisting teachers in these four areas; identifying objectives; monitoring student progress; providing feedback; and adjusting instruction (Shavelson et al., 2008). The FA professional development integrated these components across all activities.

Structure of Professional Development. There were two primary components of the FA professional development structure. The first involved a set of workshops (32 h total) that focused on building teachers' understanding and knowledge of FA followed by in-school support designed to support teachers' implementation of FA activities into their day-to-day instruction.

The 32 h of professional development were delivered: 24 h in a

three-day workshop prior to the beginning of the school year and then two 4-h after-school sessions. The first after-school session occurred two weeks into the school year and the second occurred two weeks later. Thus, the initial professional development was completed within the first month of the school year. Teachers were asked to bring a unit of instruction to the workshops that they were interested in studying in greater depth and that they would finish implementing in their classrooms in the first three months of school. These could be teacher-constructed units or those provided by publishers. Throughout the unit planning sessions, as content was presented, teachers were regularly asked to refer to their units and examine them concerning types and quality of learning targets (identifying objectives), assessments (monitoring progress), providing feedback, and progression of activities towards the learning objective (adjusting instruction). Based on what they found, teachers were asked to make adjustments, improvements, and additions to their units during the weeks between the sessions. The goal for the initial sessions was for each of the teachers to have a fully developed unit of instruction ready for implementation during the second month of school. This unit thus became the first of the three units they implemented for the project and served as a model for the remaining two units.

The in-school support was delivered through three activities: a pre-unit meeting, in class observation of implementation, and a post-unit meeting. This cycle of pre-unit, in-class, and post-unit support for grade-level teams occurred for the three units. Grade-level teams at each school selected a topic for a unit of study and then developed a pretest, a posttest, and formative assessments for the unit. Unit topics could be from any content or skill area and could change from one unit to the next. Although a few teachers focused one or more of their units on science, social studies, or writing, most focused on mathematics. Much of the pre-unit planning focused on formulating clear learning objectives and learning progressions, and then developing formative assessments to monitor students' progress toward the learning objectives. The pre-unit study also involved discussions of different strategies for using formal and informal formative assessments and how to provide feedback and adjust instruction based on information gathered from assessments. During the unit, observations were conducted in each teacher's classroom. These were scheduled in advance to occur when teachers felt they could showcase their use of FA. The post-unit sessions provided a forum for discussion where sharing related to the implementation of the unit and the observation occurred. In addition, these meetings focused on continuous improvement in unit design.

In the FA professional development, we encouraged teachers to see formative assessment as a process and involve students deeply in the process. Student involvement in formative assessment varied considerably across classrooms. In some classrooms, students worked problems on white boards and shared their thinking with the teacher and other students—students were actively monitoring learning, providing feedback and shaping the classroom activity. In others classrooms, student involvement in learning was less visible and formative assessment focused largely on assessments and assessment strategies.

1.3.2. Developing Mathematical Thinking (DMT) professional development

DMT Core Features. The DMT professional development is built around the DMT instructional model that hinges on five key dimensions of classroom practice: (a) focusing on students' initial ideas through problem solving; (b) encouraging multiple solution strategies and models; (c) pressing students conceptually; (d) maintaining a focus on the structure of the mathematics; and (e) addressing misconceptions (Brendefur, Carney et al., 2015). DMT

was chosen for this study because its multidimensional framework for student-centered teaching encourages teachers to monitor their students' mathematical ideas and to then modify their instruction based on what they observed from several perspectives. The model thus has deep but implicit expectations for teachers to conduct ongoing formative assessments as their instruction progresses, but does not include specific, focused professional development in the vocabulary and mechanics of formative assessment itself.

Structure of Professional Development. The structure and timing of the DMT professional development activities was similar to the FA activities in terms of the three-day workshop and two–four-hour after-school sessions, followed by in-class support involving three cycles of pre-unit meeting, in class observation of implementation, and a post-unit meeting.

The workshops were designed to improve teachers' mathematical knowledge for teaching (Ball, Thames, & Phelps, 2008), which involves deeply understanding: (a) mathematics topics, (b) how students' learning progresses in these topics, and (c) pedagogically how they can facilitate and promote students' understanding through the use of the DMT instructional framework. The initial three day summer workshop focused on understanding place value, multiplication, and fraction ideas through the lens of problem solving, mathematical models (e.g., bar model and number lines), the structure of number and operation (e.g., units, composing, decomposing, iterating and partitioning), and students' strategies for understanding these mathematical topics. The follow-up workshops that occurred in early fall further developed this work with an increased focus on whole and rational number operations but still through the lens of problem solving, models, structure and student thinking.

The unit study meetings were designed to assist teachers in more deeply understanding how to teach the unit (or topic) by focusing on students' thinking about the topic, the mathematics and the different possible pedagogical actions (curricular, instructional, and assessment). During the pre-unit meeting, discussions occurred around the key understandings students must hold for that topic. This was often through the lens of mathematical models and the structural components relevant to the unit of study. The post-unit meeting typically involved a discussion of what the students had and had not learned, followed by a discussion of next steps in instruction for that topic and/or unit revisions that would occur for the following year. While mathematical content of the discussion at times reiterated the information from the pre-unit study, the experience teaching the unit promoted conversations at a deeper level of reflection regarding the teachers' role and instruction and next steps in implementation. Between the pre- and post-unit studies, embedded classroom support was provided, designed to assist teachers to enact the components of the DMT instructional framework within their classroom. It was up to grade-level teams to determine the specific content focus of the lesson and then DMT staff members co-taught the lesson with the teacher with the specific pedagogical focus of how to best implement the five DMT components. Other grade level teachers observed.

The in-class lesson generally involved providing students with an open-ended mathematics task, similar to those on the pre-unit study assessment. Students were given time to engage in the task individually and encouraged to construct a mathematical model of the situation if appropriate. Then students were often asked to share their solutions with a partner or table group. This was often followed by a whole group discussion of selected solution strategies. While this was the typical model for implementation lessons, it varied greatly depending upon the interest and experience of the classroom teacher in facilitating student-led discussions. The post-unit study meeting typically focused on both the student thinking that was elicited by the task and the discussion that ensued.

1.3.3. Developing Mathematical Thinking + formative assessment (DMT + FA) professional development

DMT + FA Core Features. The DMT + FA intervention was framed on the DMT intervention with the added component of *explicit* and *focused* opportunities for teachers to formatively assess students' understanding and discuss the results from both a quantitative perspective (i.e., percentage of students proficient) and a qualitative perspective (i.e., what understanding students demonstrated).

Structure of Professional Development. The structure and timing of the DMT + FA professional development activities was similar to the FA and DMT activities in terms of the three-day workshop and two-4-hour after-school sessions, followed by in-class support involving three cycles of pre-unit meeting, in-class observation of implementation, and a post-unit meeting. The primary difference between the DMT and DMT + FA professional development was the focus on examining student responses from common assessments during the pre-unit meeting, and using this information to inform the in class implementation. Preceding each pre-unit study meeting with teachers, the DMT + FA professional development team met to: (a) determine the topics for the next unit of study; (b) identify the important understandings students must hold for that topic; and (c) create assessment items based on mathematical topics and important understandings. The expectation was that teachers would administer the assessment items to their students prior to the pre-unit study meeting. During pre-unit study meetings, teachers brought student work samples from the assessment items to examine and sort with a focus on determining what the students did and did not understand in relation to the learning objectives, what this indicated about students' informal mathematical understanding of the topic, and how this could be used to bridge students into the more formal understandings. This generally led to a discussion of next steps in instruction—for the in-class observation of implementation—both at the individual student and whole class level. The post-unit study meeting typically focused on both the student thinking that was elicited by the task and the discussion that ensued.

1.4. Measuring the efficacy of the professional development programs

The research design allowed us to evaluate whether professional development on student-centered mathematics instruction, formative assessment, or a combination of both affects absolute and relative accuracy of teachers' judgments of student learning. The Common Core State Standards has increased emphasis on conceptual understanding of mathematics; therefore, we wanted to examine judgment accuracy for both mathematical skill and conceptual understanding. Thus, the key proximal outcome variables are absolute and relative accuracy of teachers' judgment of students' mathematics performance for both skill and conceptual understanding. We hypothesized that the professional development would improve judgment accuracy.

An additional goal of this research was to examine the relationship between teachers' judgment accuracy and their students' mathematics achievement. Our theory of change is that the information gathered through monitoring informs instruction, which in turn affects student achievement. Thus, our key distal outcome variable was student achievement at the end of the academic year (controlling for performance at the beginning of the year). We hypothesized that improving teachers' judgment accuracy—either by improving use of formative assessment (FA group), or by increasing student-centered instruction of mathematics (DMT group), or by improving use of formative assessment and mathematics instruction (DMT + FA group)—would allow teachers to better tailor instruction to the individual needs of students, which

in turn would produce greater student achievement in mathematics.

2. Method

2.1. Participants and design

Two-hundred-sixty-seven elementary teachers, from grades (Kindergarten – Grade 5), participated in the study. As noted above, schools were randomly assigned to one of four conditions in a 2 (FA) $\times$ 2 (DMT) factorial design. The number of teachers in each group varied due to variations in school size. There were 65 teachers in the Control group, 60 in the FA group, 77 in the DMT group, and 65 in the DMT + FA group. All teachers predicted performance on two tests. One focused on mathematical skill and the other on conceptual understanding (i.e., kind of test was a within-participant variable). Thus, we had a 2 (FA) $\times$ 2 (DMT) $\times$ 2 (Test) design.

Ninety percent of the teachers were female ($n = 240$); gender did not differ across groups, $\chi^2(3) = 0.10$, $p = .99$. (All revised to APA format.) The distribution of teachers at each grade level did not differ across groups, $\chi^2(15) = 0.93$, $p = 1.00$. Thirty-four percent of the teachers had graduate degrees ($n = 90$); education level did not differ across groups, $\chi^2(3) = 4.15$, $p = .25$. Teachers had an average of 13.0 years of experience, which did not differ across groups, $F(3, 266) = 1.8$, $p = .16$. (Removed MSe to be consistent with APA formatting—we did this throughout the document.) We also administered a pretest of knowledge of classroom assessment (i.e., the Classroom Assessment Literacy Inventory, originally validated by [Impara, Plake, & Fager, 1993](#)) and found no difference across groups, $F(3, 259) = 1.4$, $p = .27$. Thus, the teachers in the groups were generally comparable prior to professional development.

2.2. Measures

Judgment Accuracy. We measured judgment accuracy during the spring of the academic year, after teachers had received eight months of professional development. To assess accuracy, we had teachers preview two mathematics tests taken from the Primary Mathematics Assessment ([Brendefur, Johnson, Thiede, Severson, Beaulieu, & Smith, 2015](#)). One measured math skills and the other measured conceptual understanding. Primary and intermediate grade tests were created to be age appropriate, but tests covered the same concepts. These tests were 5-item tests, and had good internal-consistency reliability (Cronbach's alpha greater than 0.80). After previewing the tests, teachers predicted the number of items each student in their class would get correct on each test. They then administered the tests to students, which were then scored by the researchers. Absolute accuracy was operationalized as confidence bias (the mean predicted performance minus the mean actual performance computed across students, as in [Griffin, Jee, & Wiley, 2009](#)). Scores closer to 0 represent more accurate judgments, positive scores occur when teachers overestimate their students' performance, and negative scores occur when teachers underestimate their students' performance. We computed a measure of absolute accuracy for both the skill and the conceptual test. Relative accuracy was operationalized as the intra-individual correlation between teachers' predictions of individual student performance and students' actual test performance computed across students (as in [Helmke & Schrader, 1987](#)). Higher correlations represent more accurate judgments. We computed a measure of relative accuracy for both the skill and the conceptual test.

Student Achievement. Student achievement was assessed using the Measures of Academic Progress® (MAP, [Northwest Evaluation Association, 2004](#)), which is a standardized test used by the

school district. The MAP was administered in the fall and spring of each academic year, starting at Grade 2. Thus, analysis of achievement data was limited to grades 2 through 5. Given the nested data structure (students within teachers) and temporal ordering of pretest and posttest data, we modeled the variation in the outcome variable using a multilevel causal mediation model.

2.3. Procedure

Teachers participated in one of four professional development groups (Control, FA, DMT, or DMT + FA). All participants provided demographic information and consented to participate in the study during an informational meeting. After eight months of professional development, teachers completed the judgment tasks.

3. Results

3.1. Predicted and actual performance on the judgment task

Judgment accuracy is the relation between predicted and actual test performance. Therefore, we report these data first. We analyzed predicted and actual performance using a 2 (FA) $\times$ 2 (DMT) $\times$ 2 (Test) analysis of variance (ANOVA), with Test as a within-participant factor. Predicted performance differed across type of test, $F(1, 263) = 216.61, p < .001, \eta_p^2 = 0.45$. As seen in Table 2, teachers predicted greater performance on the skill test than the conceptual test. Predicted performance was greater for teachers who had received FA professional development (FA and DMT + FA) than for those who had not (DMT and Control), $F(1, 263) = 7.38, p = .01, \eta_p^2 = 0.02$. Predicted performance did not differ for groups who had received DMT professional development (DMT and DMT + FA) and those who had not (FA and Control), $F(1, 263) = 2.22, p = .14$. No interactions were significant, all $F_s < 1.30$ and all $p_s > .26$.

Actual performance differed across type of test, $F(1, 263) = 183.01, p < .001, \eta_p^2 = 0.41$. As seen in Table 2, actual performance was greater on the skill test than the conceptual test. Actual performance was greater for teachers who had received DMT professional development (DMT and DMT + FA) than for those who had not (FA and Control), $F(1, 263) = 6.47, p = .01, \eta_p^2 = 0.02$. Actual performance did not differ for groups who had received FA professional development (FA and DMT + FA) than for those who had not (DMT and Control), $F(1, 263) < 1, p = .49$. No interactions were significant, all $F_s < 1$ and all $p_s > .48$.

3.2. Judgment accuracy

Absolute Judgment Accuracy. For each teacher, we computed two measures of absolute accuracy—one for the test of skill and one for the test of conceptual understanding. As noted above, absolute accuracy was operationalized as the mean predicted performance minus the mean actual performance. We examined the effect of professional development on judgment accuracy using a 2 (FA) $\times$ 2

(DMT) $\times$ 2 (Test) ANOVA.

Absolute accuracy did not differ across the type of test, $F(1, 263) < 1, p = .94$. Absolute accuracy differed for teachers who had received DMT professional development (DMT and DMT + FA) and those who had not (FA and Control), $F(1, 263) = 14.36, p = .001, \eta_p^2 = 0.05$. As seen in Fig. 1, the DMT groups were significantly less over-confident than the No-DMT groups—DMT and DMT + FA were actually under-confident. Absolute accuracy did not differ for groups who had received FA professional development (FA and DMT + FA) and those who had not (DMT and Control), $F(1, 263) = 3.81, p = .052, \eta_p^2 = 0.01$. No interactions were significant, all $F_s < 1$ and all $p_s > .35$.

To evaluate whether the main effects were consistent across the three years of the project and teachers' grade level [early elementary (K–Grade 2) versus upper elementary (Grade 3–5)], we added these factors in an ANOVA. The effects of DMT and FA were consistent across these factors. That is, year of the project did not interact with DMT or FA, $F(1, 243) < 1, p > .54$; nor did grade level, $F(1, 243) < 1.36, p > .25$.

Relative Judgment Accuracy. For each teacher, we computed two measures of relative accuracy—one for the test of skill and one for the test of conceptual understanding. Relative accuracy was operationalized as the intra-individual correlation between predicted and actual performance of the tests. As with absolute accuracy, we examined the effect of professional development on teachers' judgment accuracy using a 2 (FA: Yes versus No) $\times$ 2 (DMT: Yes versus No) $\times$ 2 (Test: Skill versus Conceptual) ANOVA.

Relative accuracy differed across type of test, $F(1, 263) = 84.39, p < .001, \eta_p^2 = 0.24$. As seen in Fig. 2, relative accuracy was greater for the test of skill than for the test of conceptual understanding. Relative accuracy also differed for teachers who had received DMT professional development (DMT and DMT + FA) and those who had

Table 2
Mean predicted and actual test performance by group.

Group	Predicted		Actual	
	Skill	Conceptual	Skill	Conceptual
DMT + FA (n = 65)	3.44 (.11)	2.80 (.11)	3.46 (.10)	2.82 (.10)
DMT (n = 77)	3.25 (.10)	2.55 (.11)	3.41 (.10)	2.79 (.10)
FA (n = 60)	3.61 (.11)	2.94 (.11)	3.28 (.10)	2.57 (.11)
Control (n = 65)	3.29 (.10)	2.75 (.10)	3.17 (.10)	2.57 (.10)

Note. The values in parentheses are the standard error of the mean.
Tables 2–4 now include sample sizes for each group.

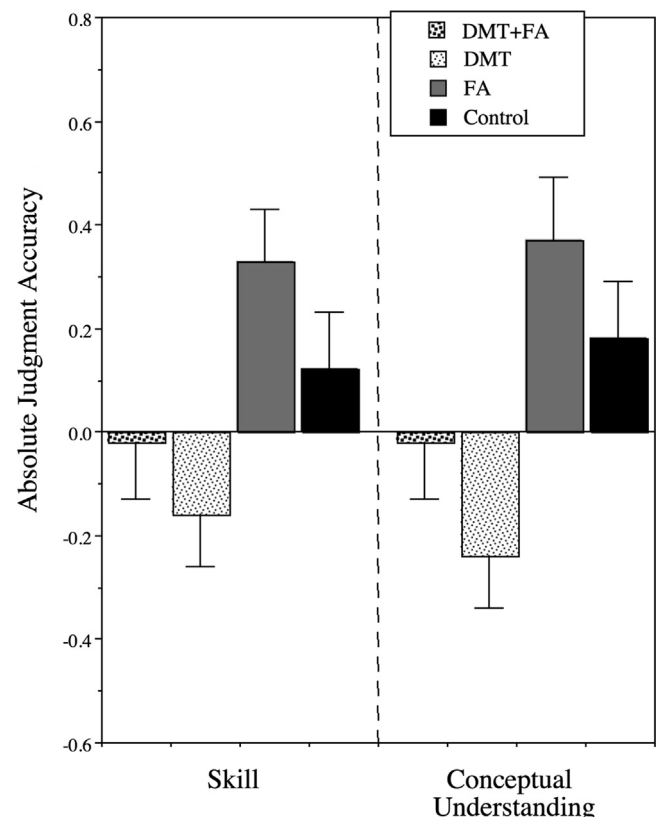


Fig. 1. Absolute judgment accuracy by test and group.

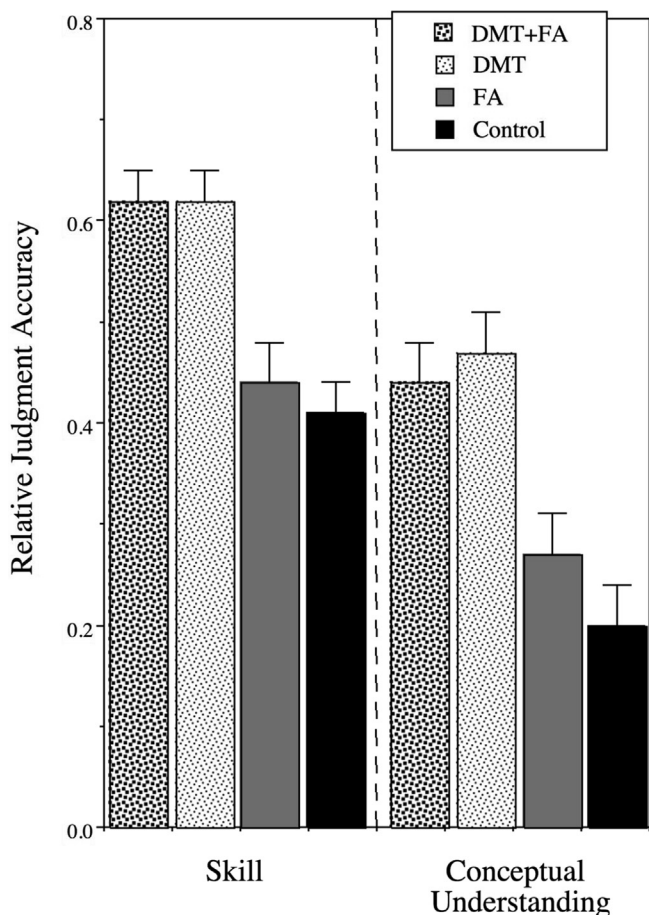


Fig. 2. Relative judgment accuracy by test and group.

not (FA and Control), $F(1, 263) = 47.21, p < .001, \eta_p^2 = 0.15$. Accuracy was greater for the DMT groups than the No-DMT groups. Relative accuracy did not differ for groups who had received FA professional development (FA and DMT + FA) and those who had not (DMT and Control), $F(1, 263) < 1, p = .57$. No interactions were significant, all $F_s < 1.35$ and all $p_s > .25$.

Again, to evaluate whether the main effects were consistent across the three years of the project and teachers' grade level [early elementary (K – Grade 2) versus upper elementary (Grade 3–5)], we added these factors in an ANOVA. The effects of DMT and FA were consistent across these factors. Year of the project did not interact with DMT or FA, $F(1, 243) < 1, p > .52$; nor did grade level, $F(1, 243) < 1, p > .40$. As noted above, we did not gather pretest data to verify judgment accuracy equivalence across groups, which could be a limitation to the current study. However, the lack of a significant interaction between DMT and Year of the Study indicates the effect of DMT was consistent across all three years of the study. Thus, although the superior accuracy for the DMT groups (DMT and DMT + FA) could be attributed to chance assignment of more accurate teachers to the DMT groups (i.e., making type I errors each year), it seems more likely the superior judgment accuracy was due to participation in the DMT professional development.

We also analyzed these data after using a Fisher-z-transformation on the intra-individual correlations. The results were consistent with those reported above. In particular, relative accuracy differed across type of test, $F(1, 263) = 71.19, p < .001, \eta_p^2 = 0.21$. As seen in Table 3, relative accuracy was greater for the test of skill than for the test of conceptual understanding. Relative

Table 3

Mean relative judgment accuracy (z_r) for the skill and conceptual test by group.

Group	Skill	Conceptual
DMT + FA (n = 65)	.89 (.07)	.56 (.06)
DMT (n = 77)	.85 (.07)	.61 (.06)
FA (n = 60)	.62 (.07)	.33 (.06)
Control (n = 65)	.48 (.07)	.22 (.06)

Note. The values are Fisher z-transformed correlations. The values in parentheses are the standard error of the mean.

accuracy also differed for teachers who had received DMT professional development (DMT and DMT + FA) and those who had not (FA and Control), $F(1, 263) = 32.22, p < .001, \eta_p^2 = 0.11$. Accuracy was greater for the DMT groups than the No-DMT groups. Relative accuracy did not differ for groups who had received FA professional development (FA and DMT + FA) and those who had not (DMT and Control), $F(1, 263) = 1.12, p = .29$. No interactions were significant, all $F_s < 1.26$ and all $p_s > .29$.

Again, to evaluate whether the main effects were consistent across the three years of the project and teachers' grade level [early elementary (K – Grade 2) versus upper elementary (Grade 3–5)], we added these factors in an ANOVA. The effects of DMT and FA were consistent across these factors. Year of the project did not interact with DMT or FA, $F(1, 243) < 1, p > .59$; nor did grade level, $F(1, 243) < 1, p > .69$.

3.3. Student achievement

Student achievement was assessed using the Measures of Academic Progress® (MAP), which was administered at the beginning and end of the school year, starting in Grade 2. Thus, only teachers in Grade 2 through Grade 5 were included in these analyses, which included 45 teachers from the DMT + FA groups, 53 teachers from the DMT group, 42 teachers from the FA group, and 44 teachers from the Control group. The MAP is reported on a scale based on Rasch analysis. Scores for our sample ranged from 135 to 270.

Student achievement data were nested within teachers; therefore, we used a multilevel regression model to examine the associations between condition and posttest achievement, while controlling for each student's pretest achievement, and clustering of students within teacher (class). Mean mathematics achievement (adjusted for differences in pretest performance) across groups is presented in Table 3.

The results of the multilevel regression model are presented in Table 4. As expected, pretest achievement was highly predictive of posttest achievement. More important, only DMT was related to student achievement, which was significantly greater for students whose teachers had participated in DMT professional development than for those whose teachers did not (Table 5).

Our theory of change is that DMT professional development improves student achievement by improving teachers' judgment accuracy, which enables teachers to make better decisions related to instructional pacing (absolute accuracy) and tailoring instruction

Table 4

Mean mathematics achievement by group.

Group	Posttest
DMT + FA (n = 65)	215.94 (.41)
DMT (n = 77)	214.85 (.38)
FA (n = 60)	213.98 (.43)
Control (n = 65)	213.31 (.42)

Note. These are adjusted mean controlling for pretest performance on the MAP test. The values in parentheses are the standard error of the mean.

Table 5
Predictors of posttest mathematics achievement.

Variables	Initial model			Mediation model		
	B	SE	z	B	SE	z
Pretest Achievement	.88	.01	118.23**	.89	.01	120.58**
DMT	1.55	.57	2.72*	-.02	.53	.04
FA	.68	.60	1.13	.31	.52	.59
DMT x FA	.41	.82	.50	.77	.71	1.07
Relative Accuracy Skill				3.61	.80	4.52**
Relative Accuracy Conceptual				1.88	.62	3.04*
Absolute Accuracy Skill				-.55	.29	1.89
Absolute Accuracy Conceptual				.15	.25	.58

Note. * indicates significance at the 0.01 level, ** indicates significance at the 0.001.

to the specific needs of individual students (relative accuracy). However, the DMT professional development may have affected student achievement through its influence on instructional factors other than judgment accuracy. Thus, we evaluated whether judgment accuracy mediated the effect of DMT on change in student achievement by adding measures of absolute judgment accuracy and relative judgment accuracy to the initial model. It is important to note that while measures of absolute accuracy for the skill and conceptual tests were moderately correlated to one another ($r = 0.61$, $p < .001$), as were measures of relative accuracy ($r = 0.58$, $p < .001$), the association between measures of absolute and relative accuracy, for each type of test, were weak and negative (ranging from $r = -0.06$ to -0.17).

As seen in the rightmost columns of Table 4 (Mediation Model), relative accuracy of skill and conceptual understanding were both predictive of posttest student achievement, whereas absolute accuracy of skill and conceptual understanding were not. More important, including relative judgment accuracy in the model eliminates the significant effects of DMT on student achievement, which suggests the DMT professional development improves student achievement by enabling teachers to make better decisions about how to tailor instruction to the needs of students.

4. Discussion

Models of instruction suggest teachers' judgments of student learning play an important role in guiding instruction (e.g., Box et al., 2015; Ready & Wright, 2011). It follows that the accuracy of teachers' judgments will influence the quality of instructional decisions, which should influence student achievement. Thus, it is important to find ways to improve judgment accuracy.

Over the past three decades, much research has examined factors that influence the accuracy of teachers' judgments of student learning (Südkamp et al., 2012), but with relatively little exploration of the cues teachers use to judge student learning. However, recent research has increasingly explored the cues teachers use to judge student learning (Machts, Kaiser, Schmidt, & Möller, 2016), such as student engagement (e.g., Kaiser, Retelsdorf, Südkamp, & Möller, 2013), student ethnicity and minority status (e.g., Kaiser, Südkamp, & Möller, 2017), and student motivation (Wijnia, Loyens, Derous, & Schmidt, 2016). Given that judgment accuracy is influenced by the cues used to judge learning (Koriat, 1997), this research is important because it helps identify the universe of cues used to judge student learning.

Based on the cue-utilization framework of monitoring (Koriat, 1997), we hypothesized that professional development that focuses teachers on cues that are diagnostic of student learning would improve judgment accuracy. We sought to evaluate whether teachers' judgment accuracy was affected by professional development on student-centered instruction and use of formative

assessment, both of which should increase the salience of cues that are diagnostic of student learning.

Student-centered teaching promotes conversations about learning among students and between students and the teacher (Connor et al., 2014; Curby, Rimm-Kaufman, & Cameron Ponitz, 2009). Thus, we hypothesized that increasing the use of student-centered approaches to teaching would increase teachers' awareness of student learning and increase the accuracy of judgments of student learning. In accord with this hypothesis, we found that both absolute accuracy and relative accuracy were greater for teachers who participated in professional development on student-centered instruction (DMT and DMT + FA) than for those who did not participate in this professional development (FA and Control). These findings are consistent with other research showing that professional development to promote student-centered approaches to teaching mathematics improves absolute judgment accuracy (e.g., Carpenter et al., 1989) and relative judgment accuracy (Thiede et al., 2015).

We attributed the improved accuracy of teacher judgment to the increased use of student-centered instruction. However, it is important to note that in addition to facilitating the development of skills in student-centered instruction, the DMT professional development also focused on increasing pedagogical content knowledge in mathematics, which is crucial for monitoring student learning (e.g., Falk, 2012). Thus, increased pedagogical content knowledge may have contributed to the improved judgment accuracy and student achievement. To effectively implement student-centered instruction, teachers need reasonable levels of mathematical pedagogical content knowledge (Ball et al., 2008). We chose the DMT professional development because it targets both of these areas. Thus, we are not able to disentangle the effects of the professional development on student-centered instruction from the effect of increasing pedagogical content knowledge. Isolating the effect of these factors on teachers' judgment accuracy and student achievement could be interesting, but we leave that to future research.

Use of formative assessment is associated with teachers' judgment accuracy—accuracy is greater when teachers more frequently use formative assessment during instruction (Thiede et al., in press). Thus, we hypothesized that increasing the use of effective formative assessment would increase the accuracy of judgments of student learning. Inconsistent with this hypothesis, we found judgment accuracy was not affected by participation in professional development on formative assessment. One explanation for this finding is that participating teachers may not have had adequate content knowledge or pedagogical content knowledge to effectively use formative assessment. This explanation is supported by research showing elementary teachers often lack adequate pedagogical content knowledge in mathematics (e.g., Ball, Hill, & Bass, 2005; Loucks-Horsley, Love, Stiles, Mundry, & Hewson, 2003), and research showing effective use of formative assessment is affected by content knowledge (Brookhart, 2011) and pedagogical content knowledge (Falk, 2012). Thus, to be effective, professional development on formative assessment may need to be delivered in a context that also supports increased content and pedagogical content knowledge (Coffey, Hammer, Levin, & Grant, 2011; Sabel, Forbes, & Zangor, 2015).

Finally, we showed that teachers more accurately judged performance on a test of skills than on tests of conceptual understanding. This is a new and important finding because it shows teachers may be better able to monitor learning and provide feedback for some kinds of learning than others. Although we did not have teachers predict performance for individual items, it may be that judgment accuracy varies across individual items as well, which may have implications for interpretation of hit rate for item-

level judgments. The current study cannot address this issue, as additional research is needed to explore whether judgment accuracy varies across domains (e.g., reading and mathematics) or items—and what factors influence accuracy across different kinds of tests, domains, or items.

Our theory of change was that improving the accuracy of teachers' judgments of student learning would enable teachers to better tailor their instruction to meet the needs of individual students, which would, in turn, increase student achievement. Teachers may struggle to effectively use their judgments of student learning to adjust instruction (Stiggins, 2008). However, when professional development focuses on students' thinking and provides additional professional development on adjusting instruction based on students' thinking within a particular content area focus, as in this study, professional development can improve student achievement.

The regression results suggest that relative judgment accuracy mediated the effect of DMT on student achievement. This suggests that although improving judgment accuracy may not be sufficient to improve student achievement, accurate judgments of student learning may be necessary to effectively guide instruction and improve student achievement. Thus, relative accuracy of teachers' judgments could be a key outcome variable for evaluating whether instructional practices improve teachers' monitoring of student learning.

Since Black and Wiliam (1998) reported the positive effect of formative assessment on student achievement, researchers have sought to design programs to improve the use of formative assessment. However, implementing a formative assessment program in a classroom can be challenging (Phelan, Choi, Vendlinki, Baker, & Herman, 2011). Moreover, evaluating the efficacy of these programs is difficult because researchers have typically engaged in extensive classroom observations to document changes in instructional practice (e.g., Ruiz-Primo & Furtak, 2006), which is very time intensive. Gathering data to examine the accuracy of teachers' judgments of student learning is a relatively low-burden way to evaluate progress toward improving monitoring of student learning, which allows researchers to isolate shortcomings in professional development and modify the program to target the areas of need. Thus, we advocate for using judgment accuracy as a proximal measure for evaluating the efficacy of professional development on formative assessment and other professional development designed to improve teachers' monitoring of student learning.

Using judgment accuracy as a proximal measure in this way adds to the growing body of research, over the past three decades, that has examined factors that influence the accuracy of teachers' judgments of student learning (Südkamp et al., 2012). The present research showed that accuracy of teachers' judgment is a malleable factor. Perhaps more important, judgment accuracy is a malleable factor linked to student achievement. Therefore, more research is needed to find ways to improve the accuracy of teachers' judgments.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at

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